UNIT 2 · TOPIC 1(A)

Power Transmission Fundamentals

Syllabus topic: Design and selection of machinery elements: gears, pulleys, chains and sprockets, belts

Power, torque and speed

$$P\ (kW) = 2\pi NT/60{,}000\ (N \text{ in rpm, } T \text{ in N·m})$$

$$P\ (hp) = 2\pi NT/4500\ (T \text{ in kgf·m})$$

Speed (velocity) ratio of any drive

$$N_1/N_2 = D_2/D_1\ (\text{belts, neglecting slip}) = T_2/T_1\ (\text{teeth — chains and gears})$$

Choosing a drive

Drive	Characteristics	Where used
Belt	Friction-dependent; slips under overload (overload relief); quiet; tolerates misalignment; cheap	Dominant on farm machines — combines, threshers
Chain	Positive (no slip); high load capacity; exact timing; needs lubrication and alignment	Planter and baler drives
Gear	Compact, positive, highest power density	Gearboxes, final drives, rotavator trains

Practice MCQs

Q1. Power transmitted by a shaft is given by:

(a) 2πNT/60 W (b) πNT/60 W
(c) NT/60 W (d) 2NT/π W

Answers

Q1 — (a) 2πNT/60 W

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 4 (Elements of Rotary Power-Transmission Systems) and Appendix (ASAE V-belt standards); standard machine design references (Shigley; V.B. Bhandari); ASAE/IS standards.